EXOTHERMIC AND ENDOTHERMIC PROCESSES

Aim: To observe some exothermic (heat releasing) and endothermic (heat absorbing) processes.

Method:

Procedure

A Solution Processes

Safety Note

- **Sodium hydroxide pellets are corrosive and must not be allowed to come in contact with your skin.**

#1 Place about 30 mL of water in a 100 mL beaker and measure its temperature. Add one teaspoonful of sodium hydroxide pellets, stir gently, and record any change in the temperature.

#2 Repeat the procedure using, in turn, a teaspoonful of ammonium chloride, sodium ethanoate and sodium chloride.

#3 Tabulate your results under the headings: solute, initial temperature, final temperature, and classify the reaction as exothermic or endothermic.

B Reaction Between Ammonium Thiocyanate and Barium Hydroxide

Safety Note

- **Barium hydroxide is corrosive and poisonous. Do not allow it to come in contact with your skin.**

#1 Place about half a teaspoonful each of solid ammonium thiocyanate and solid barium hydroxide in a test tube. Cork the test tube and shake gently until a solution is formed.

#2 Remove the cork and smell cautiously. Try to identify one of the products. Measure and record the temperature of the solution.

C Reaction Between Iron and Copper(II) Sulfate Solution

#1 Place about 25 mL of saturated copper(II) sulfate into a 100 mL beaker and record the temperature.

#2 Place a cylinder of steel wool about 1 cm thick into the solution and hold it under the surface with the thermometer. Record any evidence of reaction and change in temperature.

Results:

REACTION	OBSERVATIONS
A: Dissolving sodium hydroxide (NaOH)	
B: Reaction between ammonium thiocyanate + barium hydroxide	
C: Reaction between iron + copper sulphate	

Conclusion: Determine whether the reactions are exothermic or endothermic.